

Experiment 4

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1 Aim

In this experiment we aim to study the Boolean algebra truth tables using AND, OR, NAND, NOR and XOR Integrated Circuits (ICs) by constructing 3 logic circuits.

2 Theory

Digital electronics use discrete bands of analog levels to represent signals, unlike analog electronics, which utilize a continuous range. Within each band, all levels correspond to the same signal state. Minor variations in analog signal levels caused by manufacturing tolerances, signal attenuation, or noise remain within the designated range and are therefore disregarded by signal state sensing circuitry.

Typically, digital electronics operate with two distinct states, represented by two voltage bands: one near a reference voltage (0 V) and the other close to the supply voltage (+5 V). These states correspond to Boolean values, where 0 (OFF or **False**) and 1 (ON or **True**) define the system's logic.

In Boolean algebra, the input and output behavior of any logic gate or circuit can be systematically arranged in a table for easy visualization. This table, known as a *Truth Table*, outlines all possible input combinations along with their corresponding outputs, illustrating the logic gate's function.

2-input AND gate

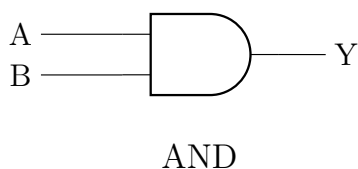
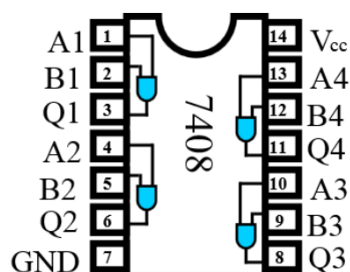


Figure 1: Logic Symbol



A	B	$Y = AB$
0	0	0
0	1	0
1	0	0
1	1	1

Figure 2: Circuit layout of AND IC

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2-input OR gate

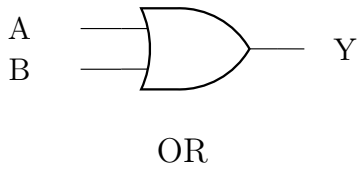


Figure 3: Logic symbol

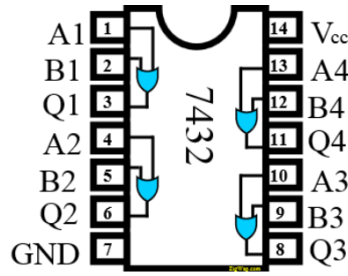


Figure 4: Circuit layout of OR IC

A	B	$Y = A + B$
0	0	0
0	1	1
1	0	1
1	1	1

2-input NOT gate

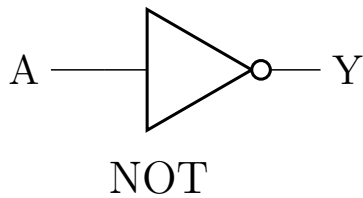


Figure 5: Logic symbol

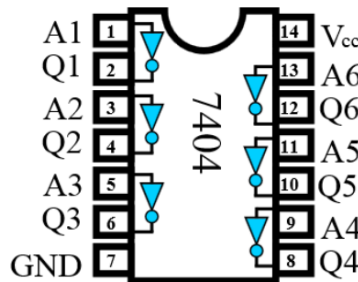


Figure 6: Circuit layout of NOT IC

A	$Y = \bar{A}$
0	1
1	0

2-input NAND gate

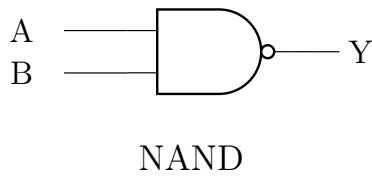


Figure 7: Logic symbol

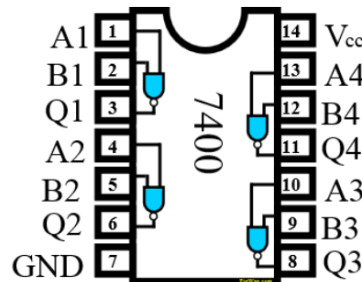


Figure 8: Circuit layout of NAND IC

A	B	$Y = \overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

2-input NOR gate

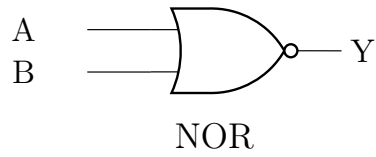
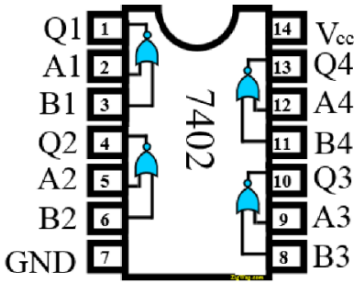


Figure 9: Logic symbol



A	B	$Y = \overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

Figure 10: Circuit layout of NOR IC

2-input XOR gate

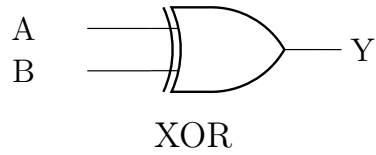
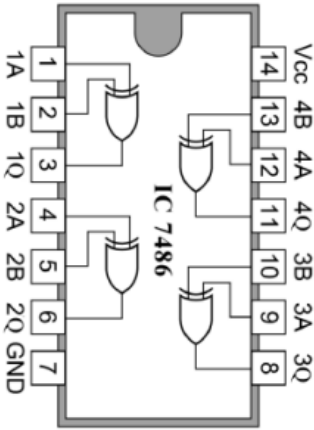


Figure 11: Logic Symbol



A	B	$Y = A \oplus B = \bar{A}B + A\bar{B}$
0	0	0
0	1	1
1	0	1
1	1	0

Table 1: Truth table for XOR gate

Figure 12: Circuit layout of XOR IC

3 Circuit Diagrams

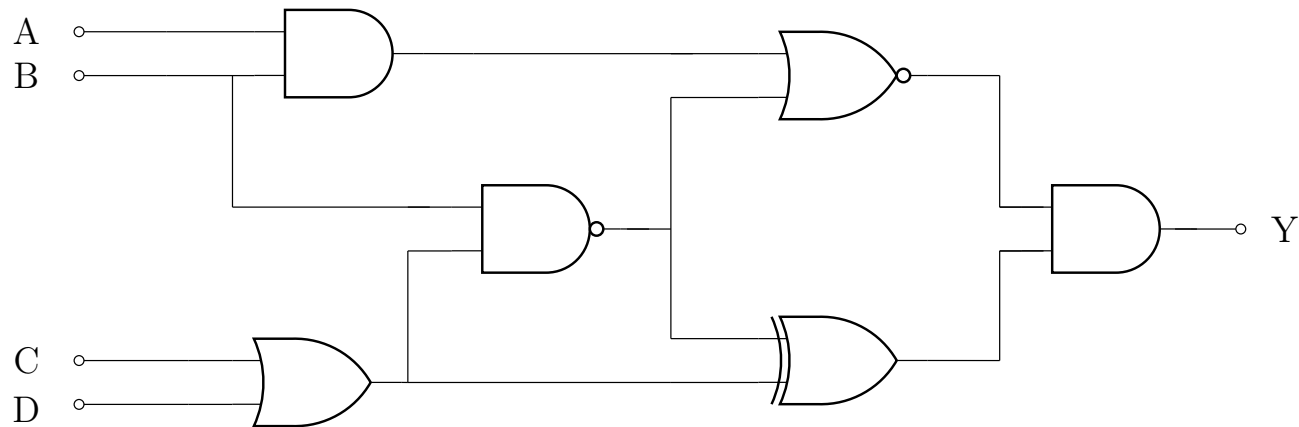


Figure 13: Logic Circuit 1 diagram

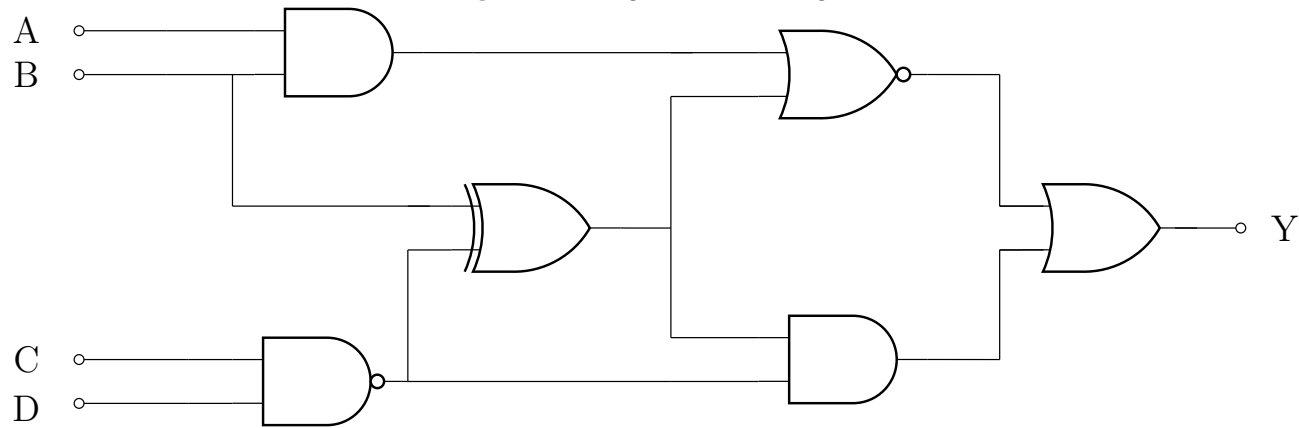


Figure 14: Logic Circuit 2 diagram

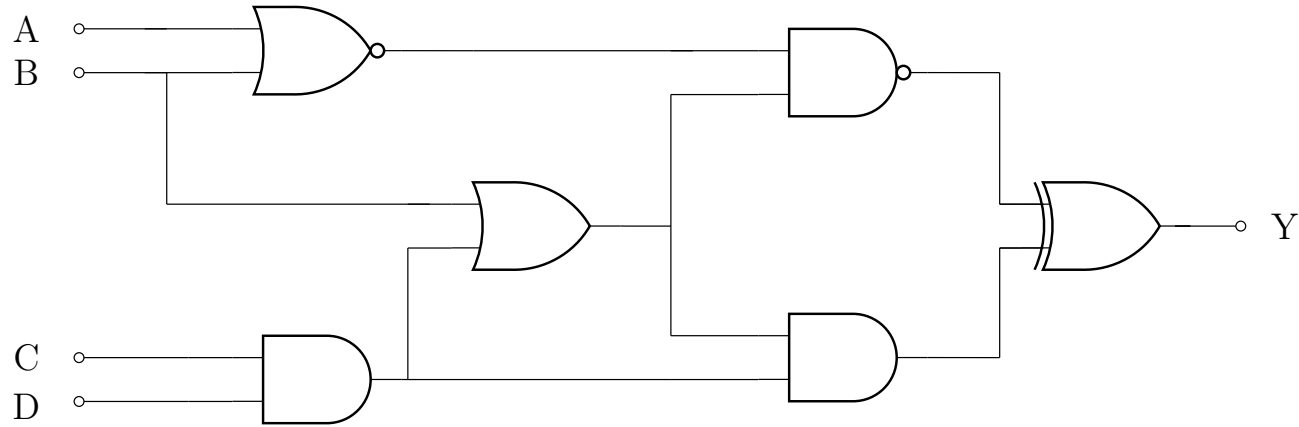


Figure 15: Logic Circuit 3 diagram

4 Truth Table

Sl No.	Inputs				Output (Y)		
	A	B	C	D	Circuit 1	Circuit 2	Circuit 3
1	0	0	0	0	0	1	1
2	1	0	0	0	0	1	1
3	0	1	0	0	0	1	1
4	0	0	1	0	0	1	1
5	0	0	0	1	0	1	1
6	1	1	0	0	0	0	1
7	1	0	1	0	0	1	1
8	1	0	0	1	0	1	1
9	0	1	1	0	1	1	1
10	0	1	0	1	1	1	1
11	0	0	1	1	0	1	1
12	1	1	1	0	0	0	1
13	1	1	0	1	0	0	1
14	1	0	1	1	0	1	0
15	0	1	1	1	1	0	0
16	1	1	1	1	0	0	0

5 Conclusion

In this experiment, we were able to learn the operations of various logic gates using Integrated circuits. All the three logic circuits were made and the truth tables were successfully verified, and **no anomalies** were observed.

6 Discussion

To make the input nodes: 1, it is connected to $+V_S$, and for 0, it is to be connected with the ground. In particular, it is to be noted that, readings due to the node dangling in air are *ambiguos* and should not be considered.

References

- B. Ghosh, Fundamental Principles of Electronics 2021. ISBN: 9788187134398